



AURORA ROBOTICS

Mobile Robot Kinematics

A Guided Textbook Chapter

Based on the AuroraMR Library

Models Covered

- Two-Wheel / Unicycle
- Ackermann / Bicycle
- Differential Drive
- Mecanum (Holonomic)

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For each model:

Derivation from first principles · Complete equation set · Step-by-step solutions ·
Numerical worked examples · Python (AuroraMR) snippets

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